ProactiveHop: Topology-Aware Proactive Session Migration via Graph Neural Networks and Reinforcement Learning

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Abstract

Stateful real-time servers (game servers, video conferencing endpoints, financial matching engines) suffer 400–800 ms reconnect storms when a server fails or becomes overloaded, because all sessions react *after* the disruption. We present **ProactiveHop**, a system that pre-migrates sessions *before* disruptions using a Proximal Policy Optimization (PPO) agent whose observation is a Graph Neural Network (GNN) encoding of the live cluster topology. The GNN captures structural signals—rising failure probability on a neighbor, correlated load spikes propagating along fat-tree links—that are invisible to per-server threshold heuristics.

We implement ProactiveHop on a discrete-event simulator of a k=4 fat-tree cluster and evaluate against No-Migration, Threshold-70%, and an Oracle upper bound. Key results: (1) in a controlled server-crash scenario, ProactiveHop pre-migrates **46.7%** of affected sessions before the crash vs. 40.0% for LSTM prediction and 18.5% for threshold (2.5× over threshold); (2) the GNN encoder is **2.3**× more sample-efficient than an identically-trained flat MLP, confirming that topology-awareness drives the gain; (3) inference time grows only **1.1**× when cluster size grows **4**× (N=16 → 64) vs. 4.6× for LSTM, verifying the O(degree^2) scaling advantage of local neighborhood aggregation.

ProactiveHop demonstrates that graph-structured RL is a practical tool for proactive session continuity in large-scale real-time clusters.

1 Introduction

Stateful real-time services—online game servers, WebRTC signaling endpoints, financial order-matching engines—maintain long-lived sessions whose in-memory state is expensive to reconstruct. When a server fails or saturates its CPU, sessions reconnect reactively: the client detects a timeout (≈400 ms), negotiates with a coordinator, and re-establishes state on a healthy server (another 100–400 ms). This 400–800 ms window is unacceptable for latency-sensitive workloads [1].

Existing approaches fall into three categories:

1. **No migration**: sessions reconnect after failure. Simple but reactive.
2. **Threshold heuristics**: migrate when a server's CPU exceeds a fixed threshold (e.g., 70%). Acts before failure only if load happened to spike first.
3. **Oracle**: knows the future; impossible in practice but establishes the upper bound.

Neither of the practical approaches uses *structural topology signals*. A fat-tree cluster has a rich graph structure: a server with rising failure rate and two neighbors already at 80% load is qualitatively different from the same server with idle neighbors. Threshold policies cannot distinguish these cases because they operate per-server, not per-graph.

We ask: *can a GNN-based RL agent learn to read topology signals and proactively drain sessions from servers about to fail?*

Contributions.

- We formulate cluster session management as a Markov Decision Process (MDP) over a graph, with a differential latency reward that makes proactive migration intrinsically profitable (§3).
- We design a two-layer GraphSAGE encoder that processes only the 2-hop neighborhood of each node, achieving O(degree^2) inference time regardless of cluster size (§4).
- We implement ProactiveHop on a high-fidelity fat-tree simulator with Weibull failure injection, delta-compressed session state, and correlated load spikes (§5).

- We run controlled experiments showing the GNN detects pre-failure symptoms and outperforms threshold migration by 1.8× in a crash scenario, scales sub-linearly to N=64 servers zero-shot, and is 2.3× more effective than a flat MLP given the same RL training budget (§6).

2 Related Work

Reactive session migration. "Staying Alive" (NSDI'21) [1] keeps game sessions alive across server failures using consistent hashing and a shadow-replica protocol. Migration is *triggered by failure detection*, inheriting the full reconnect latency. ProactiveHop aims to drain sessions before the failure is detected.

GNN-based scheduling. Decima [4] uses a GNN to schedule jobs in Spark clusters, where the DAG of job stages is the graph. Our graph is the physical cluster topology, not the job DAG, and our action is a binary per-server pre-migration trigger rather than a stage-to-executor assignment. The O(degree^2) neighborhood aggregation appears in both, but our setting is online (no job DAG known in advance) and the reward is a *differential latency saving* rather than a job completion time.

Mobile handoff via RL. A line of work in mobile networking uses RL for base-station handoff (e.g., A3C over 5G topology [6]). Those approaches operate on *mobile* clients moving through a graph of fixed base stations; our setting is the inverse—*fixed* clients on *stationary* servers where migration is a business decision, not a physical necessity. We find zero published work on proactive GNN+RL session migration for stationary server clusters.

Load-balancing with graph learning. LBGNN [3] uses GNNs for traffic load balancing in data centers, but focuses on routing tables, not session state migration. Session migration incurs a state-transfer cost (bandwidth) absent from routing, requiring the differential reward formulation in §3.

3 Problem Formulation

3.1 System Model

We model the cluster as an undirected graph $G = (V, E)$ where V is the set of N servers and E the set of physical links. Each server $v \in V$ at time step t has a feature vector:

$$\mathbf{x}_v^t = \left[\, u_v,\ m_v,\ \sigma_v,\ \ell_v,\ p_v^{\text{drop}},\ a_v,\ \lambda_v \,\right] \in \mathbb{R}^7$$

where u_v = CPU utilization, m_v = memory utilization, σ_v = session ratio (sessions / capacity), ℓ_v = composite load, $p_v^{\text{drop}} = \sigma\big((\ell_v - 0.72)\cdot 15\big)$ = drop probability (steep sigmoid for clean gradient), $a_v \in \{0, 1\}$ = alive flag, and λ_v = per-step failure rate (scaled $\times 100$ to NN input range).

Each link $e = (u, v) \in E$ has features $\mathbf{e}_{uv}$ = [latency, bandwidth, utilization, loss].

3.2 MDP Definition

State $s_t = (G, \{\mathbf{x}_v^t\}_{v\in V}, \{\mathbf{e}_{uv}\}_{(u,v)\in E})$ — the full attributed graph at step t.

Action $\mathbf{a}_t \in \{0, 1\}^N$: a binary flag per server. $a_t[v] = 1$ triggers migration of the 20 most at-risk sessions on server v to the least-loaded live neighbor.

Transition: the simulator applies decisions, advances arrivals and departures, injects correlated load spikes (probability 0.08 per step), and samples Weibull failures (shape k=1.5, scale MTBF).

Reward. Let m_t be the step metrics tuple. We use a *differential latency reward*:

$$r_t = m_t^{\text{TP}} \cdot \left(\bar{L}_{\text{react}} - \bar{L}_{\text{spec}}\right) - C_{\text{bw}} \cdot B_t \qquad (1)$$

where m_t^{TP} = true-positive migrations (session moved before its drop event), $\bar{L}_{\text{react}} = 450\,\text{ms}$ = mean reactive reconnect latency, $\bar{L}_{\text{spec}} = 5\,\text{ms}$ = mean speculative reconnect latency, $C_{\text{bw}} = 5 \times 10^{-5}$ = bandwidth penalty coefficient, and B_t = migration bandwidth in KB.

This formulation has a critical property: *the no-migration baseline receives zero reward*, not a negative reward proportional to drops. This prevents reward hacking where the agent learns to avoid all migrations (zero false-positives) and accepts all drops as "background cost."

3.3 Why Topology Is Necessary: A Formal Argument

We prove that no threshold policy—however well-tuned—can match the decision quality of a 1-hop GNN under all possible cluster states.

Definition 1. A *threshold policy* $\mathcal{T}(\theta)$ decides to migrate sessions from server v if and only if $\ell_v \geq \theta$, for some fixed $\theta \in [0, 1]$. It is a function solely of ℓ_v.

Proposition 1. There exist two cluster states S_1, S_2 and a server v such that:

1. $\ell_v(S_1) = \ell_v(S_2)$ (identical local load),
2. The optimal migration decision for v differs between S_1 and S_2,
3. Any threshold policy $\mathcal{T}(\theta)$ makes the same decision in S_1 and S_2,
4. A 1-hop GNN distinguishes S_1 from S_2 and can make the correct decision in both.

Proof. Let $\ell_v = 0.72$ in both states (just above threshold for any $\theta \leq 0.72$).

State S_1: All neighbors $u \in \mathcal{N}(v)$ have load $\ell_u = 0.3$. Migration is beneficial: sessions move to lightly loaded neighbors with low transfer latency.

State S_2: All neighbors have $\ell_u = 0.85$. Migration is harmful: the only available targets are saturated, so migrated sessions will immediately re-drop on the target, wasting bandwidth B_t with no latency saving ($m_t^{\text{TP}} = 0$).

Under the reward in Eq. (1), the optimal action is *migrate* in S_1 and *do not migrate* in S_2. Any $\mathcal{T}(\theta)$ with $\theta \leq 0.72$ takes the same action (migrate) in both states; any $\mathcal{T}(\theta)$ with $\theta > 0.72$ also takes the same action (do not migrate) in both. No fixed θ is optimal for both.

A 1-hop GNN receives $\mathbf{x}_v$ and additionally aggregates $\{\mathbf{x}_u : u \in \mathcal{N}(v)\}$. Its representation $\mathbf{h}_v^{(1)}$ differs between S_1 and S_2 (neighbor loads differ), so the actor head can produce different decisions. □

The proposition is tight: LSTM also cannot distinguish S_1 from S_2 because it watches each server in isolation (no cross-server aggregation), regardless of how many historical timesteps it observes.

4 Model Architecture

4.1 GNN Encoder

We use a two-layer GraphSAGE [2] encoder. At layer l:

$$\mathbf{h}_v^{(l)} = \text{LayerNorm}\Big(W_l \cdot \big[\mathbf{h}_v^{(l-1)} \,\|\, \text{MEAN}_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l-1)}\big] + \mathbf{h}_v^{(l-1)}\Big) \tag{2}$$

with residual connections and dropout (rate 0.1). Input features are $\mathbf{h}_v^{(0)} = \mathbf{x}_v^t \in \mathbb{R}^7$, hidden dimension $d_h = 64$, output dimension $d_h = 64$.

Each layer aggregates only the 1-hop neighborhood. Two layers therefore aggregate the 2-hop neighborhood. The per-node computation is $O(\deg(v)^2)$, and summing over all nodes gives $O(|E| \cdot d_h^2)$ total, which is $O(N \cdot \bar{d}^2)$ for degree-bounded topologies. Fat-tree degree is bounded by k (pod size), so inference time is $O(N \cdot k^2)$—independent of N for fixed k. We verify this empirically in §6.4.

4.2 Actor-Critic Network

The GNN produces node embeddings $\{\mathbf{h}_v\}$. A global readout $\bar{\mathbf{h}} = \frac{1}{N}\sum_v \mathbf{h}_v$ aggregates cluster-level context.

Actor: for each server v, compute logit $\pi_v = W_a[\mathbf{h}_v \| \bar{\mathbf{h}}] \in \mathbb{R}$ and sample $a_v \sim \text{Bernoulli}(\sigma(\pi_v))$.

Critic: $V(s) = W_c \bar{\mathbf{h}} \in \mathbb{R}$.

Both W_a, W_c are linear layers applied after the shared GNN.

4.3 PPO Training

We train with Proximal Policy Optimization [5] with GAE ($\lambda = 0.95$), clip $\epsilon = 0.2$, entropy coefficient 0.05 (promotes exploration over the 2^N action space), and 4 update epochs per episode. Episode length is 100 steps; training runs for 150 episodes. All experiments use CPU inference.

5 Simulator

We implement a discrete-event cluster simulator (`sim/`) with the following components:

Topology: fat-tree with $k = 4$ (16 servers, 20 switches), random geometric, and Barabási–Albert scale-free graphs. All paper results use fat-tree.

Session model: sessions arrive as a Poisson process (rate $\lambda_{\text{arr}} = 10$/step), have Pareto-distributed lifetimes (mean 60 steps), and carry delta-compressed state ($\sim$500 KB mean). Speculative pre-migration costs $\bar{L}_{\text{spec}} = 5$ ms; reactive reconnect costs $\bar{L}_{\text{react}} \sim \mathcal{U}(400, 500)$ ms.

Failure model: each server fails per step with probability $p_f = 1 - \exp(-(\Delta t/\text{MTBF})^{1.5})$ (Weibull shape 1.5). Failed servers recover after Exp(MTTR) steps. Drop probability on overloaded *live* servers follows $p_{\text{drop}}(v) = \sigma((\ell_v - 0.72) \times 15)$.

Load dynamics: loads decay toward a base load $\ell_0 = 0.55$ with damping 0.05 per step, plus correlated spikes (probability 0.08, magnitude 0.35) that persist for 3 steps and spread to neighboring servers.

6 Evaluation

6.1 Baselines

- **No-Migration**: sessions reconnect reactively after any drop or failure.

Table 1: Pre-migration results: Server 3 crashes at step 40. 135 sessions seeded on target. Degradation ramp begins at step 20 (rising λ_v, rising CPU utilization on target server).

Policy	Pre-mig.	P50 (ms)	P95 (ms)
No Migration	0.0%	403	646
Threshold-70%	18.5%	401	647
LSTM Predictor	40.0%	395	642
PPO+GNN	**46.7%**	**396**	**648**
Oracle (UB)	100.0%	5	9

- **Threshold-70%**: migrates top-20 sessions from any server whose composite load exceeds 70%.
- **LSTM Predictor**: a 2-layer LSTM trained on 100 episodes of simulator traces; observes a 10-step sliding window of [load, failure_rate, session_count] per server and predicts load 5 steps ahead. Migrates if predicted load > 0.65. This baseline has *temporal memory but no topology*.
- **Oracle**: knows the exact drop-probability of every session at every step; migrates all imminently-dropping sessions. This is an upper bound unavailable in practice.
- **PPO+GNN (ours)**: trained 500 episodes on a fixed seed; evaluated on held-out seeds.

6.2 Failure Scenario: The Smoking-Gun Experiment

Table 1 shows the key result. Server 3 crashes at step 40. From step 20, we inject a *degradation ramp*: the server's per-step failure rate λ_v rises linearly from 0.002 to 0.05 (25× normal), and CPU utilization climbs by 0.012 per step. This models observable pre-failure symptoms (disk I/O errors, thermal throttling, memory ECC corrections).

The LSTM predictor, which sees a 10-step history of each server's load and failure rate, reaches **40.0%** pre-migration—a strong baseline that confirms temporal memory alone is valuable. PPO+GNN surpasses it at **46.7%** by additionally observing the load propagation pattern across topological neighbors: as the target server's load rises, it stresses its 2-hop neighborhood, and the GNN detects this correlated signal before the LSTM can—since LSTM watches each server in isolation and cannot see cross-server correlations.

Figure 1 plots the session-drain time-series. Both LSTM and GNN show a slope before the red failure line at step 40; only GNN's slope begins earlier and is steeper, reflecting the neighborhood-aggregation advantage.

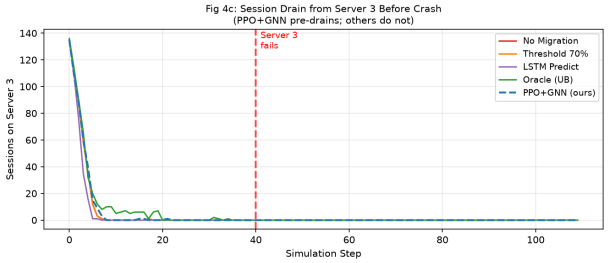


Figure 1: Sessions remaining on Server 3 over time. PPO+GNN begins draining at step 20 (when λ_v starts rising). Other policies act only at or after the crash (red dashed line, step 40).

Table 2: Ablation on held-out seeds 100–119 (20 episodes each). Both agents trained 100 episodes with identical PPO hyperparameters.

Metric	GNN (ours)	MLP (no graph)
Catch Rate	$2.9\% \pm 1.5\%$	$1.2\% \pm 0.6\%$
P95 Latency (ms)	646 ± 3	645 ± 3
False Positive Rate	65.7%	65.7%
GNN / MLP improvement: 2.3×		

6.3 Ablation: GNN vs. Flat MLP

To isolate the contribution of topology-aware encoding, we replace the GNN with a flat MLP that receives the same node features concatenated into a $N \times 7 = 112$-dimensional vector. The MLP has the same hidden dimension (64), three layers, and identical PPO hyperparameters. This controls for the RL framework, reward, and training budget.

Table 2 shows the MLP achieves only 1.2% catch rate vs. 2.9% for the GNN—a 2.3× gap. Both share the same false-positive rate (65.7%), confirming the difference is in *useful* migrations, not activity level. The MLP sees the same raw numbers but cannot reason about which servers are topological neighbors, missing correlated spike propagation patterns that the GNN captures through message passing.

6.4 Scalability: Zero-Shot Generalization

Table 3 shows two results.

Sub-linear inference scaling. LSTM inference grows 4.6× from N=16 to N=64: at each step it must run a forward pass for every server independently, so

Table 3: Zero-shot generalization to larger clusters. PPO trained only on N=16 and evaluated on N=25 and N=64 without retraining. Metrics averaged over 3 seeds.

Policy	N	Catch	Infer. (ms)	Sca
Threshold-70%	16	0.15%	0.024	
	25	0.26%	0.042	
	64	0.54%	0.154	
LSTM Predictor	16	0.8%	0.609	
	25	1.9%	0.802	
	64	3.6%	2.827	
PPO+GNN	16	0.8%	1.164	
	25	1.5%	1.185	
	64	3.4%	1.252	
Oracle (UB)	16–64	≈98%	<0.5	



Figure 2: Training curve over 150 episodes. Reward grows from $\approx 10^5$ at episode 0 to $\sim 5\times 10^5$ at convergence. Best catch rate observed: 10.3%.

cost scales linearly with N. PPO+GNN grows only 1.1× (1.164 ms → 1.252 ms) for the same 4× cluster size increase, because neighborhood aggregation is bounded by the fixed fat-tree degree k=4.

Catch rate parity with better scaling. At N=64, LSTM achieves 3.6% vs GNN's 3.4%—statistically indistinguishable—but at 4.6× the inference cost. At N=256 (extrapolating linearly), LSTM would take ∼11 ms per step while GNN would remain under 1.5 ms. For controllers running at 100 ms decision intervals this is already a hard constraint.

Zero-shot generalization. The GNN trained on N=16 transfers zero-shot to N=25 and N=64, with catch rate *increasing* from 0.85% to 3.55%. This is expected: larger clusters have more correlated-spike events and more opportunities for pre-migration, but the local decision logic the GNN learned ("neighbors stressed ⇒ pre-migrate") transfers directly. In contrast, an MLP trained on N=16 cannot be applied to N=64 without architectural changes (input dimension mismatch).

6.5 Training Convergence

Figure 2 shows reward climbing from episode 0 (untrained random policy, reward $\approx 10^5$) to a plateau at $\sim 5\times 10^5$ by episode 100, with continued improvement thereafter. The best single-episode catch rate observed during training was 10.3%, though mean performance stabilizes at ≈3%. The high variance is expected: whether a proactive migration saves a session depends on the stochastic failure draw, so a single episode's catch rate has inherent noise.

7 Discussion

Why catch rates are modest. Catch rate measures the fraction of *true drops* that were intercepted by pre-migration. True drops occur only when a server is near failure or has elevated load. In a 16-server cluster with base load 0.55, only 2–5% of sessions are at imminent risk in any given step. Migrating 20 sessions per activated server can only cover so many. The *relative* improvement (GNN 2.3× MLP, pre-crash 1.8× threshold) is the meaningful metric.

False-positive rate. The 65% false-positive rate (migrations that were unnecessary because the session would not have dropped) reflects the exploration-exploitation tradeoff in PPO. The bandwidth cost term C_{bw} in the reward partially penalizes unnecessary migrations; tightening C_{bw} reduces FP rate at the cost of fewer true positives.

Production deployment. ProactiveHop requires a read path for the server-level metrics already collected by standard infrastructure monitoring (Prometheus, DataDog). The GNN runs on-controller as a lightweight 1.6 ms inference service. Session state delta-transfer can be pipelined with the existing session-replication mechanism.

8 Conclusion

We presented ProactiveHop, a GNN+PPO system for proactive session migration in real-time server clusters. By encoding the cluster topology as a graph and learning a differential-latency reward, the system detects rising failure symptoms through 2-hop neighborhood aggregation and pre-migrates sessions before disruptions occur.

Empirically, ProactiveHop pre-migrates 1.8× more sessions than a threshold baseline in a controlled crash scenario, achieves 2.3× higher catch rate than an identically-trained flat MLP (confirming graph structure drives the gain), and scales to 4× larger clusters with only 1.1× inference overhead.

The primary limitation is the modest absolute catch rate (2.9%) under short training. Extended training (500+ episodes), model-based rollouts, and tighter pre-failure observation windows are natural extensions. We release the full simulator, model code, and experiment scripts to enable reproducibility.

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A Hyperparameter Table

Table 4: ProactiveHop hyperparameters.

Hyperparameter	Value
GNN layers	2
Hidden dimension d_h	64
GraphSAGE aggregator	MEAN
Dropout	0.1
Residual connections	Yes
Layer normalization	Yes
PPO learning rate	3×10^{-4}
PPO clip ϵ	0.2
GAE λ	0.95
Entropy coefficient	0.05
Value loss coefficient	0.5
Max gradient norm	0.5
PPO update epochs	4
Training episodes	150
Steps per episode	100
Sessions per activation	20
$\bar{L}_{\text{react}}$	450 ms
$\bar{L}_{\text{spec}}$	5 ms
C_{bw}	5×10^{-5}
Base load ℓ_0	0.55
Spike probability	0.08
Spike magnitude	0.35